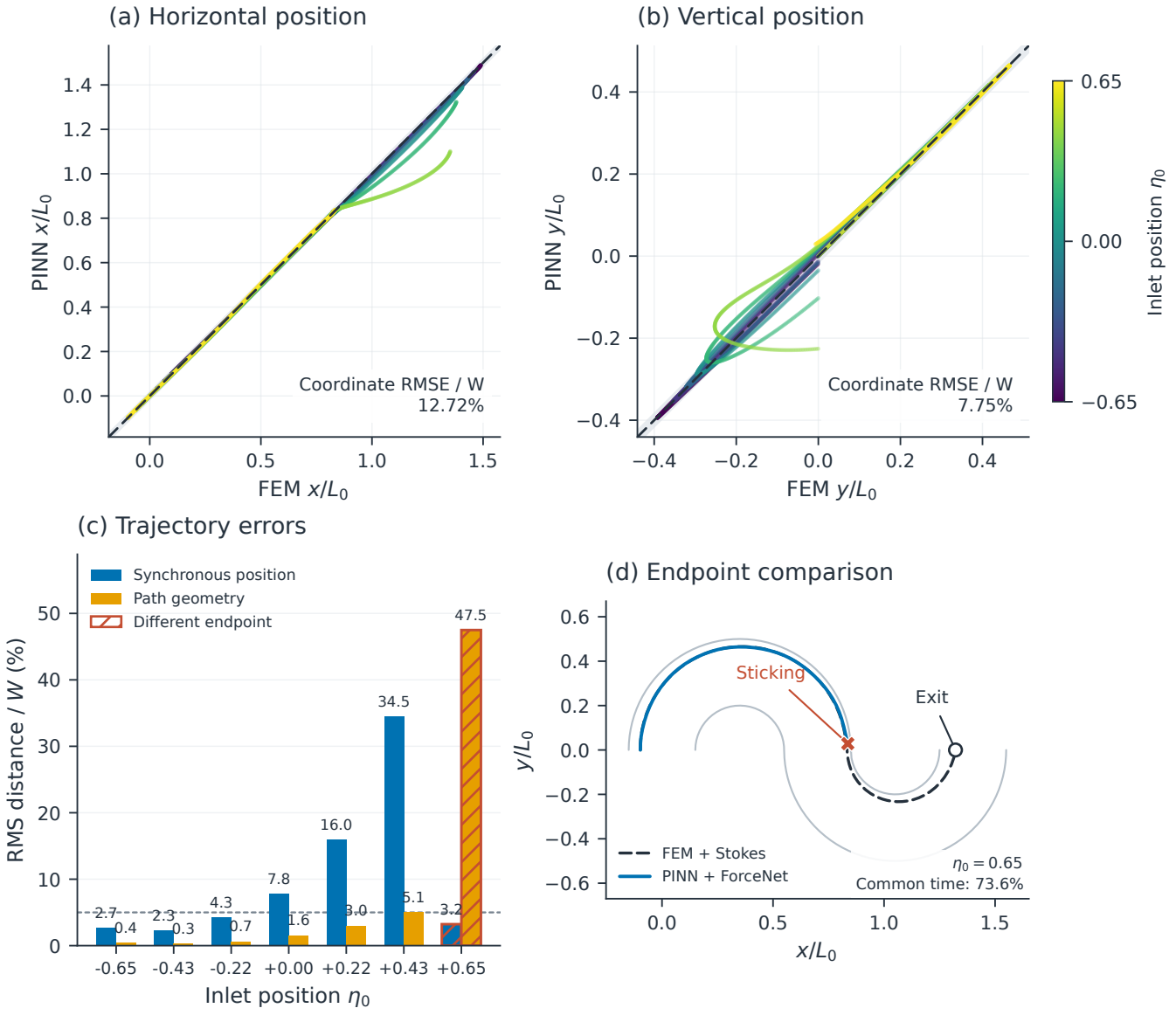

(a) Horizontal position
PINN x/L_0
FEM x/L_0
Coordinate RMSE / W
12.72%
(b) Vertical position
PINN y/L_0
FEM y/L_0
Coordinate RMSE / W
7.75%
Inlet position η_0
0.65
0.00
−0.65
(c) Trajectory errors
Synchronous position
Path geometry
Different endpoint
RMS distance / W (%)
Inlet position η_0
-0.65
-0.43
-0.22
+0.00
+0.22
+0.43
+0.65
2.7
0.4
2.3
0.3
4.3
0.7
7.8
1.6
16.0
3.0
34.5
5.1
3.2
47.5
(d) Endpoint comparison
Sticking
Exit
FEM + Stokes
PINN + ForceNet
$\eta_0 = 0.65$
Common time: 73.6%
y/L_0
x/L_0